

MLE: MVN $p(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} e^{\left\{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right\}}$

- Given n i.i.d samples drawn from $N(\mu, \Sigma)$

μ, Σ true but unknown Parameters.

- Joint likelihood of data $D = \{x_1, x_2, \dots, x_n\}$

$$F(\theta) = \prod_{i=1}^n p(x_i | \theta) = \prod_{i=1}^n \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} e^{\left\{-\frac{1}{2}(x_i - \mu)^T \Sigma^{-1}(x_i - \mu)\right\}}$$

S or Σ for covariance matrix

→ Case-I: Where μ is unknown but Σ is known

$$\text{log likelihood } F(\theta) = \sum_{i=1}^n \cdot \frac{1}{2} (x_i - \mu)^T \Sigma^{-1} (x_i - \mu)$$

$$\equiv \sum_i (x_i - \mu)^T \Sigma^{-1} (x_i - \mu) = \sum_i x_i^T \Sigma^{-1} x_i - x_i^T \Sigma^{-1} \mu - \mu^T \Sigma^{-1} x_i + \mu^T \Sigma^{-1} \mu$$

$$\equiv \sum_i -2 x_i^T \Sigma^{-1} \mu + \mu^T \Sigma^{-1} \mu$$

$$\frac{\partial F(\theta)}{\partial \theta} = \frac{\partial F(\mu)}{\partial \mu_i}$$

$$= \sum_i -2 \Sigma^{-1} x_i + 2 \Sigma^{-1} \mu = 0$$

$$x_i^T \Sigma^{-1} \mu \equiv x_i^T b$$

$$b^T x = (\Sigma^{-1} \mu)^T x$$

$$\Sigma \rightarrow d \times d \quad \mu \quad d \times 1$$

$$x_i \quad d \times 1$$

* derivative of scalar function wrt a vector

$$f(x) = x^T b = b^T x \quad b \text{ is a constant vector}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_d \end{bmatrix}$$

$$f(x) = x_1 b_1 + x_2 b_2 + \dots + x_d b_d$$

$$\text{Gradient: } \nabla_x f(x) = \begin{bmatrix} \partial f(x) / \partial x_1 \\ \partial f(x) / \partial x_2 \\ \vdots \\ \partial f(x) / \partial x_d \end{bmatrix} = b$$

$$x_i^T S^{-1} \mu = b^T \mu$$

$$\cancel{S^{-1}(x_i)} \quad \cancel{S^{-1}(x_i)} \quad (S^{-1} x_i)^T \mu$$

$$^* \nabla_{\mu} (\mu^T S^{-1} \mu) = 2 S^{-1} \mu$$

$$\text{Ans: } x^T S x = [x_1 \ x_2] \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = a_1 x_1^2 + a_2 x_2^2$$

$$\nabla_x f(x) = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \end{bmatrix} = \begin{bmatrix} 2a_1 x_1 \\ 2a_2 x_2 \end{bmatrix} = 2 \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 2 S x$$

$$\nabla_{\mu} F(\mu) = \begin{bmatrix} \partial f / \partial \mu_1 \\ \partial f / \partial \mu_2 \\ \partial f / \partial \mu_3 \end{bmatrix}$$

data matrix $X \in \mathbb{R}^{d \times N}$

$$X = \begin{bmatrix} -1 & 3 & 4 & 0 \\ 0 & 1 & -1 & 2 \\ 0 & 3 & 4 & 1 \end{bmatrix}_{3 \times 4} \quad \hat{\mu}_{MLE} = \begin{bmatrix} 1.5 \\ .5 \\ 2 \end{bmatrix}$$

$$\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i$$

⑦ We want to estimate the covariance.

$$f(\theta) = \sum_i -\frac{1}{2} \ln |S| - \frac{1}{2} (x_i - \mu)^T S^{-1} (x_i - \mu)$$

$$\frac{\partial |S|}{\partial S}$$

if a_{ij} is an element of S ,

$$\frac{\partial |S|}{\partial a_{ij}} = \frac{\partial \sum_{j=1}^n a_{ij} c_{ij}}{\partial a_{ij}}$$

$$X - \hat{\mu} = \begin{bmatrix} -2.5 & \dots \\ -.5 & \dots \\ -2 & \dots \end{bmatrix}$$

$$\sum_i b_i = b_1 + b_2 + \dots + b_i + \dots$$

$$\frac{\partial}{\partial b_i} \sum_i b_i =$$

$$\sum_j a_{ji} c_j = a_{j1} c_{j1} + \dots + a_{ji} c_j + \dots$$

$$\frac{\partial |S|}{\partial a_{ji}} = c_j$$

$$\frac{d|S|}{dS} = C = |S| S^{-1}$$

$$A^{-1} = \frac{1}{|A|} C^T$$

$$\frac{\partial \ln |S|}{\partial S} = \frac{1}{|S|} |S| S^{-1} = S^{-1}$$

$$\sum_i -\frac{1}{2} S^{-1} + S^{-1} (x_i - \mu)(x_i - \mu)^T S^{-1} = 0$$

for symmetric matrix A , $\frac{\partial}{\partial x} x^T A^{-1} x = -A^{-1} x x^T A^{-1}$

$$\sum_i -\frac{1}{2} + \frac{1}{2} (x_i - \mu)(x_i - \mu)^T S^{-1} = 0$$

$$\sum_i -\frac{S}{2} + \frac{1}{2} (x_i - \mu)(x_i - \mu)^T = 0$$

$$\begin{aligned} \hat{S}_{MLE} &= \frac{1}{n} \sum_i (x_i - \mu)(x_i - \mu)^T \\ &= \frac{1}{n} [(x_1 - \mu)(x_1 - \mu)^T + \dots] \\ &= \frac{1}{4} \begin{bmatrix} 2.5 \\ .5 \\ .2 \end{bmatrix} \begin{bmatrix} -2.5 & .5 & -.2 \end{bmatrix} + \dots \end{aligned}$$

- Binary ω_1, ω_2 $P(\lambda|\omega_1) \sim N(\mu_1, \Sigma_1)$ $P(\lambda|\omega_2) \sim N(\mu_2, \Sigma_2)$

$$\hat{\mu}_1, \hat{\mu}_2, \hat{\Sigma}_1, \hat{\Sigma}_2$$

data matrix for $\omega_1 \rightarrow X_1$
" " $\omega_2 \rightarrow X_2$

⑦ $X_i \sim U(0, \theta)$ $X_1, X_2, \dots, X_n$

$$P(X) = \begin{cases} 1/\theta & 0 \leq X \leq \theta \\ 0, & \text{otherwise} \end{cases}$$

$$F(\theta) = ? = \prod_{i=1}^n P(X_i | \theta) = \theta^{-n}$$



$x_{\max}$ as the max of $\{x_1, x_2, \dots, x_n\}$

$$\hat{\theta}_{MLE} = x_{\max}$$

⊗ Discrete r.v. Bernoulli r.v.

$$p(x) = \theta^x (1-\theta)^{1-x}$$

$$F(\theta) = \ln \prod_{j=1}^n \theta^{x_j} (1-\theta)^{1-x_j} = \sum_j x_j \ln \theta + (1-x_j) \ln(1-\theta)$$

$$\hat{\theta}_{MLE} = \frac{1}{n} \sum_j x_j$$

data matrix $X = \begin{bmatrix} 1 & -1 & 0 & 1 & 2 \end{bmatrix}$

④ If x is a vector & each element x_j (j^{th} element)

is a Bernoulli r.v.

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_j \\ \vdots \\ x_d \end{bmatrix} \quad \begin{aligned} P(x_j = 1 | \theta_j) &= \theta_j^{x_j} (1 - \theta_j)^{1 - x_j} \\ P(x_j | \theta_j) &= \end{aligned} \quad \begin{aligned} P(x_j = 1 | \theta_j) &= \theta_j \\ P(x_j = 0 | \theta_j) &= 1 - \theta_j \end{aligned}$$

$$P(x | \theta) = \prod_{j=1}^d \theta_j^{x_j} (1 - \theta_j)^{1 - x_j}$$

$$F(\theta) = \prod_{j=1}^n P(x_j | \theta) = \prod_{j=1}^n \prod_{j=1}^d \theta_j^{x_j} (1 - \theta_j)^{1 - x_j}$$

$$\hat{\theta}_j \text{ MLE}$$

⑦ M.V.G $\mu \rightarrow d$ params $\Sigma \rightarrow d^2$ $d(d+1)/2$

If d is large, say 1000

⑧ Bias of the estimate.

An estimator will be biased if, say for mean,

$\mu \rightarrow$ true mean, $\hat{\mu} \rightarrow$ estimated

$$E(\hat{\mu}) \neq \mu$$

$$E(\hat{\mu}) = \mu \rightarrow \text{unbiased}$$

⑨ For the Gaussian case

$$x_i \sim N(\mu, \sigma^2)$$

$$\begin{aligned} E\left(\frac{1}{n} \sum_i x_i\right) &= \frac{1}{n} \sum_i E(x_i) \\ &= \frac{1}{n} \sum_i \mu = \mu \end{aligned}$$

The MLE estimate of mean for Gaussian is unbiased

⑦ MLE estimate of variance for Gaussian

$$E(\hat{\sigma}^2) = \sigma^2? \quad \checkmark \quad \begin{array}{l} \text{unbiased} \\ \text{biased} \end{array}$$

$$E(\hat{\sigma}^2) = \frac{n-1}{n} \sigma^2 \quad \text{MLE underestimated the var. by a factor of } \frac{n-1}{n}$$

Unbiased estimate of var. $E\left\{\frac{1}{n-1} \sum_i (x_i - \hat{\mu})^2\right\} = \sigma^2 \checkmark$

⊗ MLE Bernoulli r.v. | Coin toss: H, H, H

$$\hat{\theta}_{MLE} = \frac{1}{n} \sum_i x_i$$

$$\hat{\theta}_{MLE} = 1, \quad \text{Prob. of observing } H = 1$$

$$P(x|\theta) \quad \text{Prior}(\theta)$$

$$P_{\text{posterior}} \propto P(x|\theta) P(\theta) \equiv P(\theta|x)$$

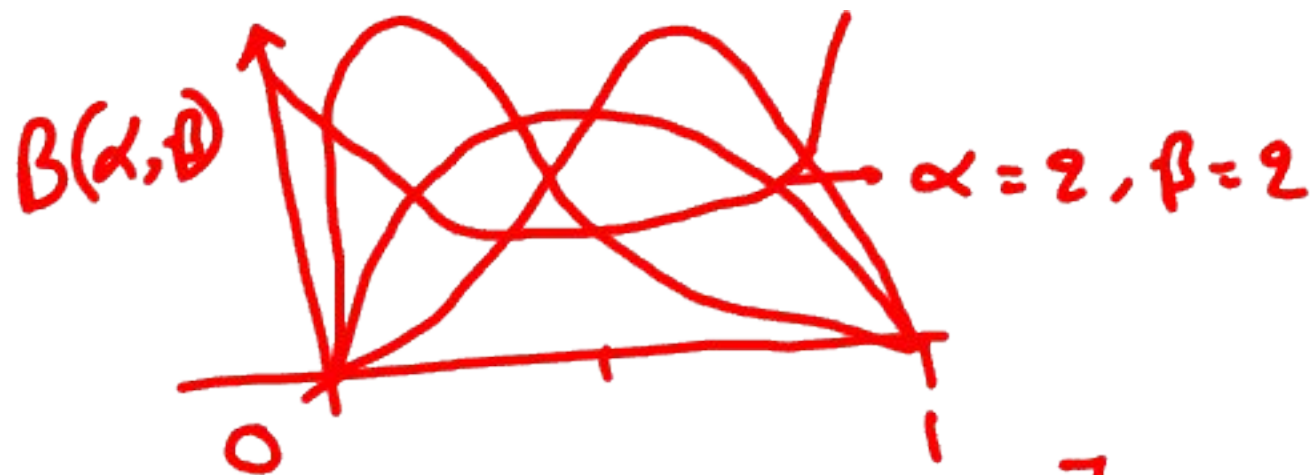
$$\underset{\theta}{\text{argmax}} P(\theta|x) = \hat{\theta}_{MAP}$$

MAP $\rightarrow$ Maximum
a-posterior

$$\max_{\theta} p(\theta|x) \equiv \max_{\theta} p(x|\theta) p(\theta)$$

- ⑧ Conjugate Priors $\rightarrow$ Posteriors take the family of distribution as Priors
 - ⑧ Bernoulli likelihood $\rightarrow$ Conjugate Priors as Beta-distr.
Gaussian $\rightarrow$ Gaussian.
 - ⑧ Beta distr. $p(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$
- α, β are parameters of beta distr.

$$\alpha = \beta = 2$$



$$P(x|\theta) P(\theta) = \left[\prod_{i=1}^n \theta^{x_i} (1-\theta)^{1-x_i} \right] P(\theta)$$

$$= \left[\sum_{i=1}^n x_i \ln \theta + (1-x_i) \ln(1-\theta) \right] + (\alpha-1) \ln \theta + (\beta-1) \ln(1-\theta)$$

$$\hat{\theta}_{MAP} = \frac{\sum_i x_i + \alpha - 1}{N + \alpha + \beta - 2}$$

$$f(\alpha, \beta) = \frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

$$x_1 = x_2 = x_3 = 1, \quad \hat{\theta}_{MLE} = 1 \quad \alpha = \beta = 5$$

$$\hat{\theta}_{MAP} = \frac{3 + \alpha - 1}{3 + \alpha + \beta - 2} = \frac{3 + 4}{3 + 8} = \frac{7}{11}$$

$$p(\theta) p(\theta | x) = \frac{\hat{\theta}_{MAP}^{\alpha-1} (1 - \hat{\theta}_{MAP})^{\beta-1}}{f(\alpha, \beta)}$$

If we take MLE, $p(x|\theta) = \prod_{i=1}^n (\hat{\theta}_{MLE})^{x_i} (1 - \hat{\theta}_{MLE})^{1-x_i}$

$$MAP, p(x|\theta) = \prod_{i=1}^n (\hat{\theta}_{MAP})^{x_i} (1 - \hat{\theta}_{MAP})^{1-x_i}$$

⑧ $\hat{\mu}_{MAP}$ for Gaussian.

$$P(x|\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(\frac{x_i - \mu}{\sigma})^2}$$

$$P(\mu; \underbrace{\mu_0, \sigma_0^2}_{\text{Parameters}}) = \frac{1}{\sqrt{2\pi}\sigma_0} e^{-\frac{1}{2}(\frac{\mu - \mu_0}{\sigma_0})^2}$$

var. $\downarrow$

Prior is on μ with mean μ_0 , var σ_0^2

$$\left(\sum_{i=1}^n \left[-\frac{1}{2\sigma^2} (x_i - \mu)^2 \right] - \frac{1}{2\sigma_0^2} (\mu - \mu_0)^2 \right)$$

take derivative w.r.t μ & equate to zero
that gives $\hat{\mu}_{MAP}$

$$\text{likelihood} = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{\left\{-\frac{1}{2}\frac{(x_i-\mu)^2}{\sigma^2}\right\}}$$

$$P(\theta|x) \propto P(x|\theta) P(\theta)$$

$$\ln P(x|\theta) + \ln P(\theta)$$

$$\sum_{i=1}^n -\frac{1}{2\sigma^2} (x_i - \mu)^2 +$$

④ Supervised setting/learning.

Dataset $\mathcal{D} = \{x_i, y_i\}_{i=1}^N$ x_i are observed samples $x_i \in \mathbb{R}^d$
 y_i are labels for each x_i

for a classification setting $y_i \in \{0, 1, 2, \dots, C-1\}$
 C - classes, N - total number of samples.

train set, val set, test set.

$$\text{train} \cap \text{val} = \emptyset$$

$$\text{train} \cap \text{test} = \emptyset$$

$$\text{val} \cap \text{test} = \emptyset$$

train set will be used to learn my parameters.
for instance discriminant is of the form $W^T x + b$
 W, b are params obtained from train set.

test set is to test the model's performance.

Measurement of performance $\rightarrow$ Accuracy.

$$\text{Acc.} = \frac{\text{Total no. of correct predictions}}{\text{Total no. of test samples.}}$$

⑦ Unsupervised setting.
 $\mathcal{D} = \{x_i\}_1^N$ $y \notin X$, no. of classes $\rightarrow$ not available.

⑧ LDA/QDA $\rightarrow$ supervised setting.

⑨ PCA [Bishop's Ch.12 latent continuous variables]
Principal Component Analysis.